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Product Review Sentiment Analysis

Course Project Report

Submitted by

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ABSTRACT

Online retailers such as Amazon collect customer reviews far faster than any team could read them, which is why automated sentiment analysis has become a standard tool for understanding customer opinion. The base paper for this project approached the problem with Bag-of-Words and TF-IDF features and a set of classical classifiers, including Naive Bayes, Linear SVC and Random Forest. Because these features treat a review as an unordered collection of words, they struggle with negations such as "not good" and with mixed reviews such as "bad camera but great battery". The base system also gives a single label to each review, depends on a separate model for emotion, and does not explain its decisions.

This project replaces that pipeline with a context-aware model built on a fine-tuned DistilBERT transformer. A single multi-task network predicts the overall sentiment of a review (positive, neutral or negative), its dominant emotion (joy, sadness, anger, fear, disgust or surprise) and the sentiment expressed toward individual product aspects, while Integrated Gradients shows which words drove each prediction. The system works on a single review typed by a user as well as on large review files, and returns either a result card or a labelled file with a summary dashboard.

The model was trained and evaluated on 113,569 reviews from the Cell Phones and Accessories category of Amazon Reviews 2023, supported by the GoEmotions and SemEval-2014 datasets. On a common test set of 11,357 reviews it reached an accuracy of 76.1% and a macro-F1 score of 0.764, compared with 68.3% and 0.684 for the strongest base paper model, Logistic Regression. It also achieved a macro-F1 score of 0.700 for emotion detection and 0.712 for aspect-level sentiment, and processed roughly 146 reviews per second on a single NVIDIA T4 GPU.

Keywords: sentiment analysis, emotion detection, aspect-based opinion mining, transformer, BERT, explainable AI, Amazon product reviews

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1. INTRODUCTION

E-commerce platforms such as Amazon receive an enormous number of customer reviews every day. Taken together, these reviews describe what buyers value, what disappoints them and how they feel about a product, which makes them one of the most useful sources of feedback a business has. The difficulty is scale: no team can read them all, so the opinions they contain have to be extracted automatically.

Sentiment analysis, also called opinion mining, is the branch of Natural Language Processing (NLP) concerned with deciding whether a piece of text expresses a positive, negative or neutral opinion [5], [6]. The base paper, "Sentimental Analysis on Amazon Product Review" [1], built a system of this kind. It cleaned the review text, converted it into Bag-of-Words and TF-IDF features, and trained five classical models: Naive Bayes, Linear SVC, Logistic Regression, Multilayer Perceptron (MLP) and Random Forest. Emotions such as joy, sadness, anger and disgust were detected by a separate model.

Although this approach is simple and fast, it has clear weaknesses. Word-frequency features discard word order, so a negation such as "not good" or a contrast introduced by "but" is easily misread. The base paper's own results illustrate the problem: the review "Unreliable connectivity, frequently experiences dropped calls" was labelled Very Positive. Because the system assigns one label to the whole review, the detail in a mixed review such as "bad camera, but great battery" is lost. It also offers no reason for its decisions and can only process a fixed CSV file.

This project keeps the base paper as its reference baseline and builds on it with the following contributions:

1. **A context-aware transformer encoder.** DistilBERT replaces the TF-IDF features, so each review is interpreted as a sentence rather than as a bag of words.
2. **A single multi-task model.** Sentiment and emotion are predicted together by one network instead of two separate models.
3. **Aspect-level sentiment.** Each product feature mentioned in a review, such as the camera, battery or price, receives its own polarity.
4. **Explainable predictions.** The words that most influenced each result are highlighted for the user.
5. **Two input modes.** The system accepts a single review entered by the user or a large file containing hundreds of thousands of reviews.

The remainder of the report is organised as follows. Section 2 reviews related work, and Section 3 describes the proposed system and its four modules. Section 4 presents the datasets, the implementation and the results of each module. Section 5 compares the proposed system with the base paper's methods, and Section 6 concludes with directions for future work.

2. LITERATURE SURVEY

The 35 works reviewed in this section fall into three groups: classical machine learning for review sentiment, deep learning and transformer models, and aspect-level analysis together with explainable AI. Each group is discussed in turn, followed by a summary of the gaps that this project addresses.

2.1 Sentiment Analysis of Product Reviews Using Classical Machine Learning

Pang, Lee and Vaithyanathan [2] were among the first to treat sentiment as a text classification problem. Training Naive Bayes, Maximum Entropy and SVM classifiers on movie reviews with unigram features, they reached about 83% accuracy, and they observed that sentiment is harder to classify than topic because a handful of words, negations in particular, can reverse the meaning of a sentence. Turney [3] took an unsupervised route instead, scoring phrases by their association with the words "excellent" and "poor" and averaging those scores to decide whether a review recommended a product.

Hu and Liu [4] mined customer reviews of electronic products, extracting frequently mentioned product features and summarising the opinion expressed about each one. Their work is an early form of the aspect-level analysis on which the present project builds. Pang and Lee [5] and Liu [6] later produced detailed surveys of the field, describing the main levels of analysis (document, sentence and aspect) and its long-standing open problems, among them negation, sarcasm and domain dependence. Medhat et al. [7] surveyed sentiment algorithms more broadly and grouped them into machine learning, lexicon-based and hybrid approaches.

Fang and Zhan [8] worked with about 5.1 million Amazon reviews and proposed categorisation at both the sentence and the review level. Comparing Naive Bayes, Random Forest and SVM, they found that Random Forest performed best at the sentence level, while SVM and Naive Bayes performed best at the review level. Parihar and Bhagyanidhi [9] reviewed sentiment analysis methods for product reviews.

The base paper [1] follows this classical pipeline on Amazon reviews. Its preprocessing consists of tokenisation, stop-word removal and lemmatisation, after which TF-IDF features are used to compare Linear SVC, Multinomial Naive Bayes, Logistic Regression, MLP and Random Forest. Gope et al. [10] compared machine learning and deep learning models for classifying the sentiment of Amazon product reviews, and Arora et al. [11] evaluated deep learning classifiers on product reviews, noting that such models capture context better but require large amounts of labelled data.

Observation: classical models are fast and easy to build, but Bag-of-Words and TF-IDF features ignore word order and context. As a result, they make systematic errors on negation, contrast and mixed opinions.

2.2 Deep Learning and Transformer Models for Sentiment and Emotion

Kim [12] showed that a simple Convolutional Neural Network (CNN) built on pre-trained word vectors performs well on sentence-level sentiment tasks. Socher et al. [13] introduced the Stanford Sentiment Treebank together with a Recursive Neural Tensor Network, which handled negation and contrast by composing phrase meanings bottom-up through the parse tree. In their survey of deep learning for sentiment analysis, Zhang, Wang and Liu [14] covered CNN, RNN, LSTM and attention models and concluded that deep models capture context better, although they need more labelled data.

The Transformer of Vaswani et al. [15] replaced recurrence with self-attention, allowing every word to attend directly to every other word and making long-range context far easier to capture. Devlin et al. [16] built BERT on this architecture: the model is pre-trained on a large corpus with masked language modelling and then fine-tuned for tasks such as sentiment classification, where it set new state-of-the-art results. Liu et al. [17] refined BERT's pre-training to produce RoBERTa, and Sanh et al. [18] compressed BERT into DistilBERT, which retains about 97% of its language understanding while being 40% smaller and 60% faster. Sun et al. [19] studied how best to fine-tune BERT for text classification and showed how the learning rate, the choice of layer and the handling of long texts affect accuracy.

For emotion, Ekman [20] proposed six basic emotions (anger, disgust, fear, joy, sadness and surprise), and these are adopted as the emotion classes in this project. Demszky et al. [21] released GoEmotions, a set of 58,000 Reddit comments labelled with 27 emotions plus neutral, together with a mapping to Ekman's six categories; it is used here to train the emotion head.

Caruana [22] introduced multi-task learning, in which a single model learns several related tasks at once and each task benefits from a shared representation. Liu et al. [23] applied the idea to NLP with MT-DNN, a shared BERT encoder with task-specific output layers, and reported better generalisation than single-task models.

Observation: transformer models read words in context and largely resolve the negation problem, and multi-task learning allows one model to predict several related outputs. None of the product-review studies reviewed here combines sentiment and emotion in a single shared transformer.

2.3 Aspect-Level Sentiment Analysis and Explainable AI

Schouten and Frasincar [24] surveyed aspect-level sentiment analysis and divided it into two steps: identifying the aspects being discussed (for example, the camera or the battery) and classifying the sentiment expressed toward each one. Pontiki et al. [25] organised SemEval-2014 Task 4, which provided labelled laptop and restaurant reviews for aspect term extraction and aspect polarity, and which has since become the standard benchmark for the task.

Tang, Qin and Liu [26] proposed a deep memory network that applies several attention layers over the context to judge the sentiment toward a given aspect. Wang et al. [27] developed an attention-based LSTM in which the aspect is added to the input, so that the model concentrates on the words related to it. Sun, Huang and Qiu [28] recast aspect sentiment as a sentence-pair task for BERT by constructing an auxiliary question such as "what do you think of the battery?", and Xu et al. [29] post-trained BERT on review text and reported large gains on both aspect extraction and aspect sentiment.

On the explainability side, Ribeiro, Singh and Guestrin [30] proposed LIME, which explains any classifier by fitting a simple local model around an individual prediction. Lundberg and Lee [31] proposed SHAP, which draws on game theory to give each input feature a fair share of the prediction. Sundararajan, Taly and Yan [32] introduced Integrated Gradients, a method for deep networks that produces word-level importance scores and satisfies desirable properties such as sensitivity and completeness. Jain and Wallace [33] showed that raw attention weights do not always explain a model's decision, which is a strong argument for using a gradient-based method rather than attention alone.

Finally, two datasets are central to this area. Hou et al. [34] released Amazon Reviews 2023, which contains more than 570 million reviews across 33 categories, and McAuley and Leskovec [35] released the Amazon Fine Food Reviews dataset of 568,454 reviews, which was also used in the base paper.

2.4 Research Gaps

Table 2.1 summarises the main gaps in existing work and the way in which the proposed system responds to each of them.

Table 2.1: Research gaps in existing work and how the proposed system addresses them.

Gap in existing work	How the proposed system addresses it
TF-IDF and Bag-of-Words ignore context and negation [1], [8]	Fine-tuned transformer encoder [16], [18]
Sentiment and emotion are predicted by separate models [1]	One shared encoder with two heads (multi-task) [22], [23]

Gap in existing work	How the proposed system addresses it
Most product-review systems give one label for the whole review [1], [10]	Aspect extraction with per-aspect polarity [24], [28]
Predictions come with no explanation [1]	Word-level Integrated Gradients scores [32]
Systems take only a fixed file as input [1]	Single-review mode and large-file batch mode

3. PROPOSED SYSTEM

The proposed system takes review text as input, interprets it with a fine-tuned transformer, and returns the sentiment, emotion and aspect-wise opinion it finds, together with an explanation of each result. It supports two input modes and is organised into four modules, as shown in Figure 3.1.

Input modes. In single-review mode, the user types one review into the web interface and receives a result card in about a second. In bulk mode, the user uploads a CSV file that may contain anywhere from 100,000 to 1,000,000 or more reviews; the file is read in chunks, processed in batches, and returned as a labelled file accompanied by a dashboard.

Processing. After cleaning in Module 1, each review is passed to a shared DistilBERT encoder. The encoder's output feeds a sentiment head and an emotion head that are trained together (Module 2), and the same encoder is used to judge the sentiment expressed toward each product aspect found in the review (Module 3). Finally, Integrated Gradients scores the contribution of every word, and the results are presented as a result card or a dashboard (Module 4).

Training. The model is trained once, offline. Sentiment labels are derived from star ratings: one or two stars are treated as Negative, three stars as Neutral, and four or five stars as Positive. Emotion labels come from GoEmotions [21], mapped to Ekman's six emotions [20], and aspect polarity labels come from the SemEval-2014 laptop reviews [25]. All three heads are trained together by minimising the combined loss $L = L_{sentiment} + \lambda \cdot L_{emotion} + \mu \cdot L_{aspect}$, where $\lambda = 0.5$ and $\mu = 0.5$ balance the contribution of each task.

System Architecture

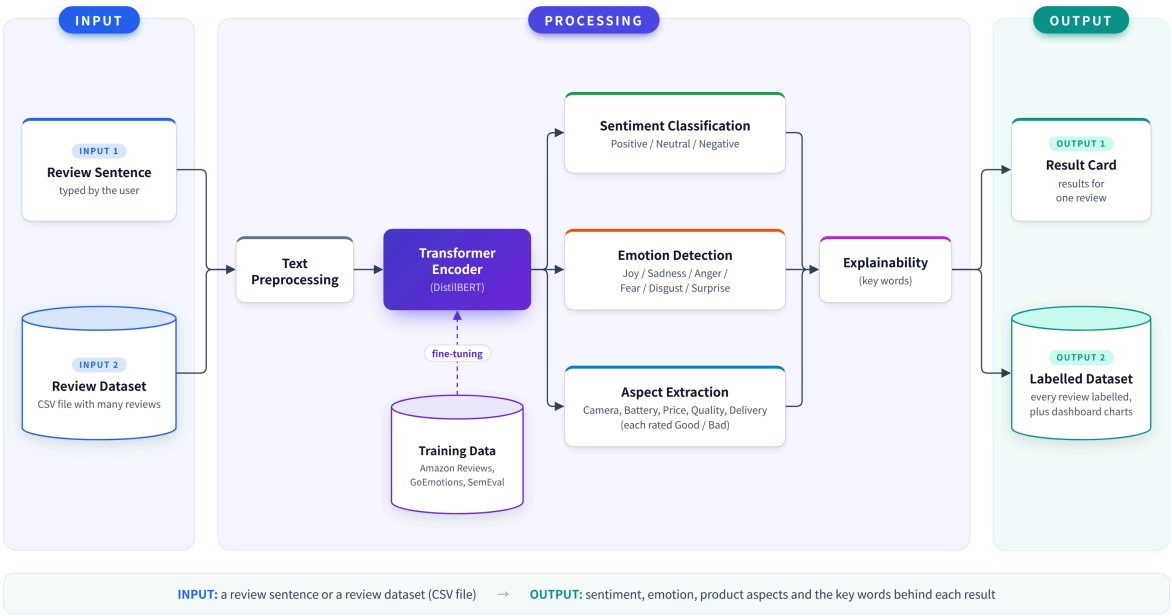


Figure 3.1: Block diagram of the proposed system. Input: a review typed by the user, or a dataset file containing many reviews. Output: sentiment (positive, neutral or negative), emotion, product aspects with their opinion, and the key words behind each result, presented as a result card or as a labelled dataset with dashboard charts.

3.1 Module 1: Data Input and Preprocessing

This module accepts either a single sentence or a large file and turns it into clean, tokenised batches for the model. For a file, it first checks that the required columns are present (review text, star rating, product ID and date) and then reads the data in chunks of 50,000 rows, which keeps memory use low even for a file of 1,000,000 reviews. Empty rows, duplicate reviews, HTML tags, URLs and very short reviews (fewer than three words) are removed. Unlike the base paper, the module deliberately does **not** remove stop-words such as "not" and "but", since the transformer relies on them to recognise negation and contrast. The text is then tokenised with the transformer's own tokenizer and padded or truncated to a fixed length of 128 tokens.

Input: one review, or a CSV file. **Output:** batches of token IDs and attention masks (with labels during training).

```
ALGORITHM 1: Data Input and Preprocessing
INPUT : source  (a sentence string OR a file path)
        mode    ("single" or "bulk"), chunk_size = 50000, max_len = 128
OUTPUT: stream of batches (input_ids, attention_mask, row_ids)

BEGIN
  IF mode = "single" THEN
      texts <- [source]
      YIELD Tokenize(Clean(texts), max_len)
  ELSE
      Check file has columns {review_text, rating, product_id, date}
      FOR EACH chunk IN ReadInChunks(source, chunk_size) DO
          chunk <- DropEmptyRows(chunk)
          chunk <- DropDuplicates(chunk, key = review_text)
          FOR EACH row IN chunk DO
              t <- RemoveHTMLandURLs(row.review_text)
              t <- NormaliseSpaces(t)          // stop-words are KEPT
              IF WordCount(t) < 3 THEN skip row
          END FOR
          FOR EACH batch OF batch_size rows IN chunk DO
              YIELD Tokenize(batch.texts, max_len), batch.row_ids
          END FOR
      END FOR
  END IF
END

FUNCTION RatingToLabel(r)            // used only in training
  IF r <= 2 THEN RETURN Negative
  ELSE IF r = 3 THEN RETURN Neutral
  ELSE RETURN Positive
```

3.2 Module 2: Multi-Task Transformer for Sentiment and Emotion

This module is the core contribution of the project. A single fine-tuned DistilBERT encoder reads each review and produces a 768-dimensional vector for the [CLS] token, which summarises the whole sentence. Two small output layers sit on top of this shared vector: the sentiment head predicts Negative, Neutral or Positive, and the emotion head predicts one of Ekman's six emotions [20]. Because both heads are trained together, what the encoder learns for one task can support the other [22], [23]. Together, they replace both the TF-IDF features and the two separate models of the base paper.

One practical difficulty is that no single dataset carries both kinds of label: the review data has sentiment labels but no emotion labels, while GoEmotions has the reverse. Training therefore alternates between batches from the two sources, and each batch updates only the loss for the label it carries.

Input: batches from Module 1. **Output:** a sentiment label and confidence, an emotion label and confidence, and token embeddings for Module 3.

```
ALGORITHM 2: Multi-Task Sentiment and Emotion Model
INPUT : review batches R (with sentiment labels),
        emotion batches E (GoEmotions, with emotion labels),
        epochs = 2, lr = 2e-5, lambda = 0.5
OUTPUT: trained model M = (Encoder, SentHead, EmoHead)

// ---------- Training ----------
BEGIN
  Encoder  <- LoadPretrained("distilbert-base-uncased")
  SentHead <- Linear(768 -> 3)
  EmoHead  <- Linear(768 -> 6)
  FOR epoch = 1 TO epochs DO
      FOR EACH batch b IN Interleave(R, E) DO
          H   <- Encoder(b.input_ids, b.attention_mask)
          cls <- Dropout(H[:, 0, :])                 // [CLS] vector
          loss <- 0
          IF b has sentiment labels THEN
              loss <- loss + CrossEntropy(SentHead(cls), b.sentiment)
          IF b has emotion labels THEN
              loss <- loss + lambda * CrossEntropy(EmoHead(cls), b.emotion)
          Backpropagate(loss); AdamW_Step(lr)
      END FOR
      Evaluate macro-F1 on validation set; keep best model
  END FOR
  RETURN M
END

// ---------- Prediction ----------
FUNCTION Predict(M, batch)
  H   <- M.Encoder(batch)
  cls <- H[:, 0, :]
  ps  <- Softmax(M.SentHead(cls));  pe <- Softmax(M.EmoHead(cls))
  RETURN (argmax ps, max ps), (argmax pe, max pe), H
```

3.3 Module 3: Aspect Extraction and Aspect-Level Sentiment

This module identifies the product parts a review talks about and assigns each one its own sentiment, following the two-step approach described by Schouten and Frasincar [24]. In the first step, candidate aspects are taken from the review's noun phrases using spaCy and kept only if they appear in an aspect list built from the most frequent nouns in the training reviews of that category (for example case, screen, price, charger and battery). Variants of the same word are grouped, so that "cam" and "camera" count as a single aspect.

In the second step, the review and each aspect are passed to the encoder as a sentence pair, for example "Bad camera, but great battery [SEP] battery", following the auxiliary-sentence method of Sun et al. [28]. A third output head on the shared encoder predicts the polarity toward that aspect, and this head is fine-tuned on the SemEval-2014 laptop reviews [25]. In bulk mode, the aspect results from all reviews are aggregated to show which aspects customers praise most and which they criticise most.

Input: clean review text. **Output:** a list of (aspect, polarity, confidence) for each review, and an aspect summary per product.

```
ALGORITHM 3: Aspect Extraction and Aspect Polarity
INPUT : review text t, trained encoder M.Encoder, aspect head AspHead,
        aspect list A (frequent nouns per category), synonym map S
OUTPUT: list of (aspect, polarity, confidence)

BEGIN
  results <- []
  doc <- spaCy_Parse(t)
  candidates <- {}
  FOR EACH noun_chunk nc IN doc DO
      head <- FirstNounInList(nc, A, S)     // nouns scanned left to right
      head <- S.get(head, head)              // "cam" -> "camera"
      IF head IN A THEN candidates.add(head)
  END FOR
  FOR EACH aspect a IN candidates DO
      x   <- Tokenize(t, a)                  // pair: [CLS] t [SEP] a [SEP]
      cls <- M.Encoder(x)[:, 0, :]
      p   <- Softmax(AspHead(cls))           // Neg / Neu / Pos
      results.append((a, argmax p, max p))
  END FOR
  RETURN results
END

FUNCTION ProductAspectSummary(all_results, product_id)   // bulk mode
  counts <- Count(aspect, polarity) over reviews of product_id
  top_praised    <- aspects sorted by Positive share, top 5
  top_complained <- aspects sorted by Negative share, top 5
  RETURN counts, top_praised, top_complained
```

3.4 Module 4: Explanation and Output

The final module explains why the model reached its answer and presents the results. For each prediction, Integrated Gradients [32] measures how strongly each word pushed the model toward the predicted label. It does so by moving gradually from an empty (padding) input to the actual input and accumulating the gradients along the way. Integrated Gradients was chosen over raw attention weights because attention alone is not a reliable explanation [33]. The three highest-scoring words are highlighted for the user.

In single-review mode, the output is a result card showing the sentiment, emotion, aspects and highlighted words. In bulk mode, every row of the uploaded file receives new columns (sentiment, emotion, aspects and confidence), and a dashboard summarises the sentiment and emotion shares, the most frequently mentioned aspects and the share of positive reviews over time. Because word-level explanations are relatively expensive to compute, bulk mode produces them only for a small sample of reviews.

Input: model outputs from Modules 2 and 3. **Output:** a result card (single-review mode) or a labelled file and dashboard (bulk mode).

```
ALGORITHM 4: Explanation and Output Generation
INPUT : text t, model M, predicted class c, steps m = 50, mode
OUTPUT: top-k important words; result card OR labelled file + dashboard

FUNCTION IntegratedGradients(t, M, c, m)
  x  <- Embeddings(Tokenize(t))
  x0 <- Embeddings(PaddingTokens(len(x)))      // baseline input
  total_grad <- 0
  FOR k = 1 TO m DO
      xk <- x0 + (k / m) * (x - x0)
      total_grad <- total_grad + Gradient(M.score(xk, c), xk)
  END FOR
  attr <- (x - x0) * total_grad / m
  word_scores <- SumOverEmbeddingDim(attr), merged per whole word
  RETURN Top(word_scores, k = 3)

BEGIN
  IF mode = "single" THEN
      (s, e, H) <- Predict(M, t);  aspects <- AspectModule(t)
      words <- IntegratedGradients(t, M, s.label, m)
      SHOW ResultCard(s, e, aspects, words)
  ELSE
      FOR EACH batch IN Module1(file) DO
          append Predict + AspectModule outputs as new columns
      END FOR
      sample <- small random sample of rows; add IG explanations to sample
      SAVE labelled_file
      SHOW Dashboard(sentiment %, emotion %, top aspects,
                     positive share over time)
  END IF
END
```

4. RESULTS AND IMPLEMENTATION

4.1 Dataset Description

Three public datasets are used in this work. The Amazon reviews are used to train and test the sentiment head and the bulk-file mode, GoEmotions is used to train the emotion head, and SemEval-2014 is used to train the aspect polarity head. Table 4.1 summarises the three datasets.

Table 4.1: Datasets used in this project.

Dataset	Source	Total size	Used in this project	Labels	Used for
Amazon Reviews 2023 [34]	McAuley Lab, UCSD	570M+ reviews, 33 categories	Cell Phones and Accessories: 120,000 reviews sampled (40,000 per class), 113,569 after cleaning	Star rating 1–5 → Neg / Neu / Pos	Sentiment training, large-file test
GoEmotions [21]	Google Research	58,009 Reddit comments	35,170 comments that map to one of the 6 Ekman emotions	27 emotions + neutral	Emotion head training
SemEval-2014 Task 4, Laptops [25]	SemEval	About 3,800 sentences	2,313 labelled aspect pairs; 15% held out for testing, since the public test split has no labels	Aspect terms + polarity	Aspect polarity head training

The Amazon reviews are split into 80% for training, 10% for validation and 10% for testing, with the class ratio preserved in each split. The base paper's models are re-run on exactly the same test split so that the comparison is fair.

Implementation environment. The system is implemented in Python as a single Google Colab notebook running on a free NVIDIA T4 GPU with 16 GB of memory. A complete run, from streaming the data to producing the final comparison, takes about 40 minutes. The software used is listed in Table 4.2, and the training settings are given in Table 4.3.

Table 4.2: Implementation environment.

Item	Used in this project
Language and platform	Python 3, Google Colab, NVIDIA T4 GPU
Deep learning	PyTorch, Hugging Face Transformers, Hugging Face Datasets
Classical ML (baseline)	scikit-learn (TF-IDF, Naive Bayes, Linear SVC, Logistic Regression, MLP, Random Forest), NLTK

Item	Used in this project
Aspect extraction	spaCy (en_core_web_sm)
Explainability	Captum (Integrated Gradients)
Charts	Matplotlib
Web interface	Gradio (public share link from Colab)

Table 4.3: Training settings of the proposed model.

Setting	Value
Pre-trained model	distilbert-base-uncased
Maximum review length	128 tokens
Batch size	32
Epochs	2
Optimizer, learning rate	AdamW, 2e-5, weight decay 0.01, 6% warm-up
Loss weights	λ (emotion) = 0.5, μ (aspect) = 0.5, one aspect batch every 16 steps
Reviews per sentiment class	40,000 (balanced)
Random seed	42

4.2 Module 1: Data Input and Preprocessing

Module 1 streams the Amazon Reviews 2023 data directly from Hugging Face, so the 9.34 GB category file never has to be downloaded. It keeps up to 40,000 reviews per sentiment class, removes empty rows, duplicates, HTML tags, links and reviews shorter than three words, and maps star ratings to sentiment labels. The same functions are reused in bulk mode, where an uploaded CSV file is read in chunks of 50,000 rows.

In total, the module retained 113,569 clean reviews out of the 120,000 sampled. Collecting 40,000 reviews per class required streaming 500,577 reviews from the category file, which took 109 seconds, and the cleaning step itself ran at roughly 15,000 reviews per second. Table 4.4 shows how many reviews remained after each cleaning step, and Table 4.5 gives the resulting splits.

Table 4.4: Reviews remaining after each cleaning step.

Cleaning step	Reviews left
Reviews read (40,000 per class)	120,000
After removing empty rows	120,000
After removing duplicate reviews	115,551

Cleaning step	Reviews left
After removing reviews under 3 words	113,569

Table 4.5: Training, validation and test splits with the number of reviews in each class.

Split	Reviews	Negative	Neutral	Positive
Training	90,855	31,013	30,739	29,103
Validation	11,357	3,876	3,843	3,638
Test	11,357	3,877	3,842	3,638

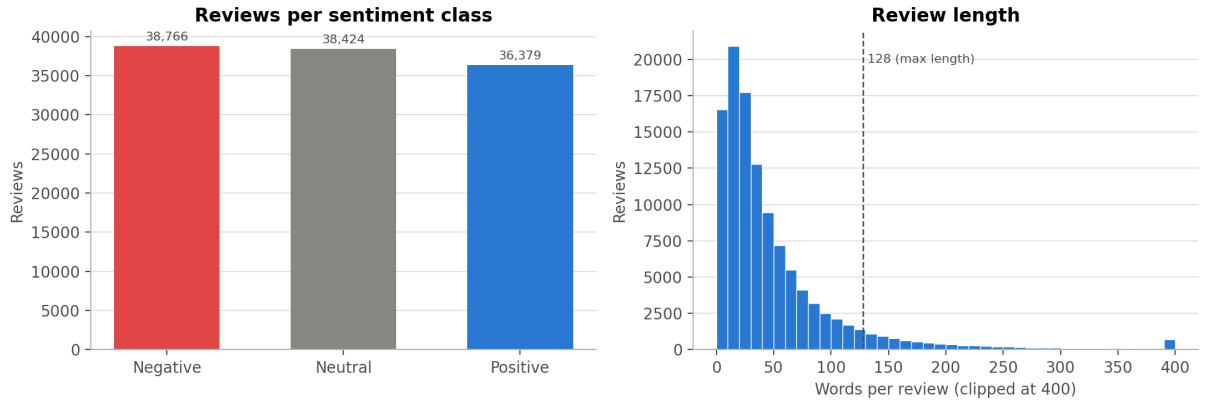


Figure 4.1: Reviews per sentiment class after cleaning (left) and number of words per review (right).

Inference. Duplicate reviews were the main data-quality problem, accounting for 4,449 reviews (3.7%), compared with 1,982 reviews that were too short to keep. After cleaning, the three classes remain close in size (36,379 to 38,766 reviews), so no single class dominates training and accuracy can be read alongside macro-F1 without much distortion. Most reviews are short, with a median length of 31 words, and only 7.5% exceed 128 words; the 128-token limit therefore truncates only a small share of the data.

4.3 Module 2: Multi-Task Transformer for Sentiment and Emotion

Module 2 fine-tunes a single DistilBERT encoder with three output heads. At each training step the model sees one batch of reviews (sentiment loss) and one batch of GoEmotions comments (emotion loss, weighted to compensate for rare emotions), and every 16 steps it also sees one batch of SemEval aspect pairs (aspect loss). The model is evaluated on the validation set after every epoch, and the epoch with the best sentiment macro-F1 is kept. To find out whether multi-task learning actually helps sentiment, a second DistilBERT model was trained on sentiment alone with identical settings.

On the 11,357 test reviews, the multi-task model achieved an accuracy of 76.1% and a macro-F1 score of 0.764 for sentiment. On the 3,539 GoEmotions test comments, it reached 80.7% accuracy and a macro-F1 score of 0.700 for emotion. Training took 17.6 minutes for two epochs on the T4 GPU.

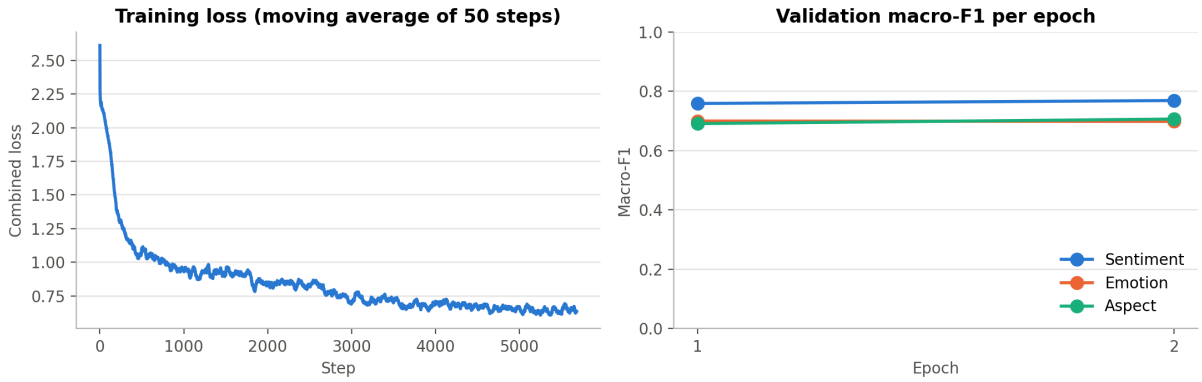


Figure 4.2: Training loss (left) and validation macro-F1 of the three heads after each epoch (right).

Inference. The combined loss falls steadily from about 2.2 to 0.65, which shows that all three heads were learning. Between the first and second epochs, validation macro-F1 improved for sentiment (0.759 to 0.769) and for aspect polarity (0.691 to 0.706), while emotion stayed essentially flat (0.700 to 0.698), so the model from epoch 2 was retained.

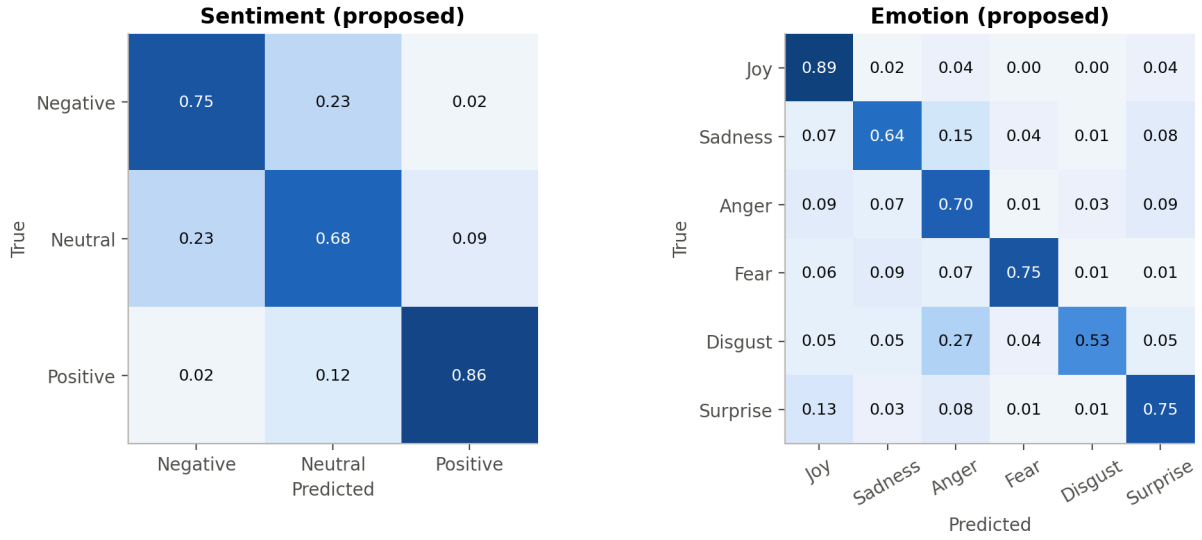


Figure 4.3: Confusion matrices for sentiment (Amazon test set) and emotion (GoEmotions test set), shown as the share of each true class.

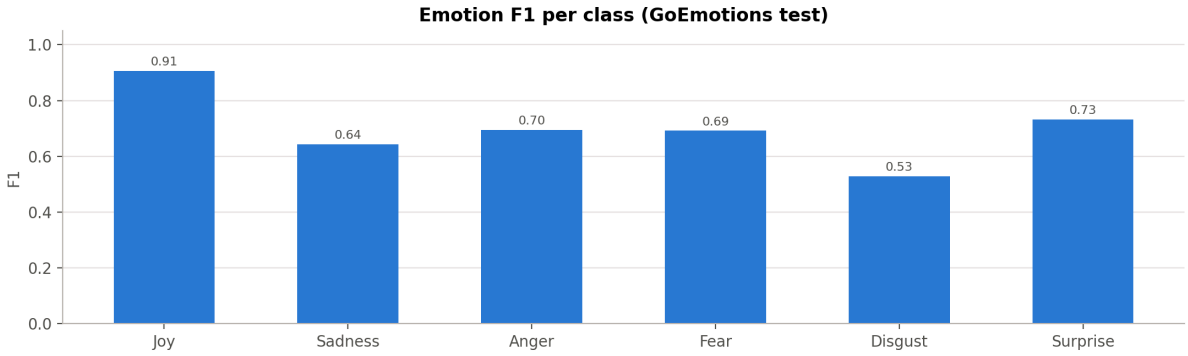


Figure 4.4: F1 score for each emotion on the GoEmotions test set.

Inference. Positive reviews are recognised most reliably, with 86% classified correctly. Neutral is clearly the hardest class (68%): almost a quarter of neutral reviews (23%) are predicted as negative, which is understandable given that three-star reviews often contain complaints. Positive and negative reviews, on the other hand, are almost never confused with each other (2%). Among the emotions, joy scores highest (F1 0.91), largely because it has the most training examples (15,809 of 28,104). Disgust scores lowest (F1 0.53); it has only 524 training examples, and 27% of disgust comments are predicted as anger, a closely related emotion. The class weights appear to help, since fear still reaches an F1 of 0.69 with only 566 examples.

Table 4.6 compares the multi-task model with the sentiment-only model trained under the same conditions.

Table 4.6: Multi-task model compared with a sentiment-only model on the 11,357 test reviews.

Model	Accuracy	Precision	Recall	Macro-F1	Training time	Outputs
DistilBERT, sentiment only	0.766	0.768	0.767	0.768	9.9 min	Sentiment
Multi-task DistilBERT (proposed)	0.761	0.765	0.763	0.764	17.6 min	Sentiment, emotion, aspect polarity

Inference. Adding the emotion and aspect tasks did not improve sentiment classification: the multi-task model scores about 0.4 points below the sentiment-only model, a difference small enough to be negligible in practice. The value of the multi-task design lies elsewhere. A single 66-million-parameter encoder now produces all three outputs, whereas the alternative would be a separate model for each task.

4.4 Module 3: Aspect Extraction and Aspect-Level Sentiment

Module 3 builds a list of the 25 most frequent product nouns from the training reviews, extracts aspects from noun phrases with spaCy, and passes each (review, aspect) pair to the aspect head. Its accuracy was measured on SemEval-2014 laptop pairs held out from training and compared with two simpler alternatives: running the sentiment head on the clause that contains the aspect, and simply assigning the whole-review sentiment to every aspect.

The aspect head reached 75.8% accuracy and a macro-F1 score of 0.712 on the 347 held-out pairs, roughly 20 points of macro-F1 above both alternatives, as Table 4.7 shows.

Table 4.7: Aspect polarity results on 347 held-out SemEval-2014 aspect pairs.

Method (SemEval-2014 laptops, 347 aspect pairs)	Accuracy	Macro-F1	Neutral F1
Aspect head, sentence pair (proposed)	0.758	0.712	0.520
Sentiment head on the clause containing the aspect	0.588	0.500	0.113
Whole-review sentiment for every aspect	0.591	0.508	0.154

Inference. A single label for the whole review cannot describe the opinion about one particular part. The dedicated aspect head raises macro-F1 from 0.508 to 0.712, and its largest gain is on neutral aspects (F1 0.52 against 0.15), which are exactly the cases where the opinion about one part differs from the tone of the sentence as a whole.

On the Amazon data, the 25 aspect words learned from the training reviews include case, screen, protector, price, charger, cable, color, battery and camera. In a sample of 3,000 test reviews, 68.4% mention at least one of these aspects, producing 3,971 review-aspect pairs (1.32 per review) that were processed in 14.3 seconds.

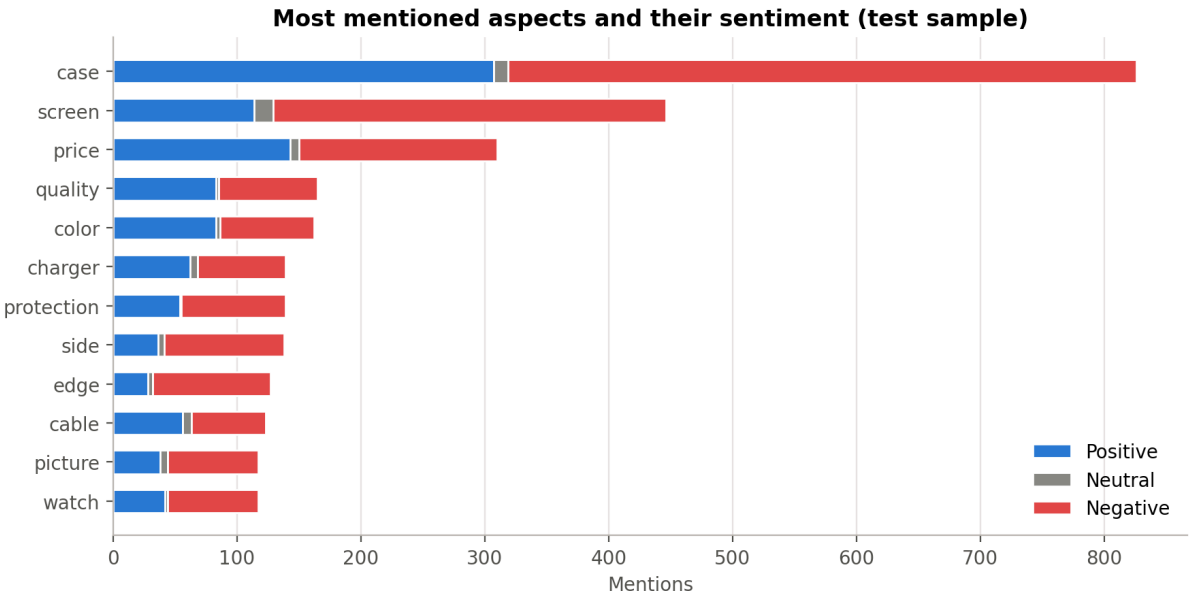


Figure 4.5: The 12 most mentioned aspects in 3,000 test reviews, split by predicted sentiment.

Inference. Because this category consists mainly of accessories, accessory parts dominate the list, and battery and camera rank only 20th and 22nd. Since the sample contains equal numbers of negative, neutral and positive reviews, the chart is best read as showing which parts attract complaints relative to the others. Cases are mentioned most often (826 times, 61% of them negative). Edges (75% negative) and screens (71% negative) attract the most complaints, which points to screen protectors that do not fit properly at the edges. Price, color, quality, charger and cable are split close to evenly.

Table 4.8 shows a few test reviews together with the aspect sentiment identified by the model.

Table 4.8: Example test reviews with the aspect sentiment identified by the model.

Review (shortened)	Aspect sentiment found
The case fits well, provides great protection and works flawlessly with my wireless charging stand. Looks and feels good. High quality phone case...	case: Positive, protection: Positive, quality: Positive
This thing is garbage... The glass doesn't match up, no matter what you do you leave a bubble on the edge.	glass: Negative, edge: Negative
Super sturdy and protective case... I probably have too many cards in it yet it still closes.	case: Positive, card: Positive
The rubber stretches over time. Ok case. Doesnt protect screen very much.	case: Negative, screen: Negative

4.5 Module 4: Explanation and Output

Module 4 uses Integrated Gradients with 50 integration steps to score each word's effect on the prediction and reports the three most important words. It produces the result card in single-review mode and, in bulk mode, a labelled CSV file and a dashboard. To compare the proposed model with the best base paper model on difficult language, a set of 14 hard sentences was prepared, covering negation, contrast and sarcasm and including the review that the base paper itself labelled incorrectly.

Besides explaining each prediction, the model correctly classified 12 of the 14 hard sentences, against 8 for the best base paper model, and labelled the full test file at 146 reviews per second. Table 4.9 shows the result card produced for a typical mixed review.

Table 4.9: Result card for the sentence "Bad camera quality, but the battery life is great."

Field	Output
Sentiment	Neutral (confidence 0.80)
Emotion	Joy (confidence 0.30)
Aspects	camera: Negative, battery: Negative
Key words	great, but, battery

Inference. Neutral is a reasonable overall label for a review that contains one complaint and one compliment, and the camera is correctly marked as negative. The battery, however, should be positive: the aspect head, which was trained on only 1,769 laptop pairs, carried the negative tone of the opening clause over to the second clause. The emotion confidence of 0.30 is also low, which suggests that the emotion head is uncertain on mixed reviews of this kind.

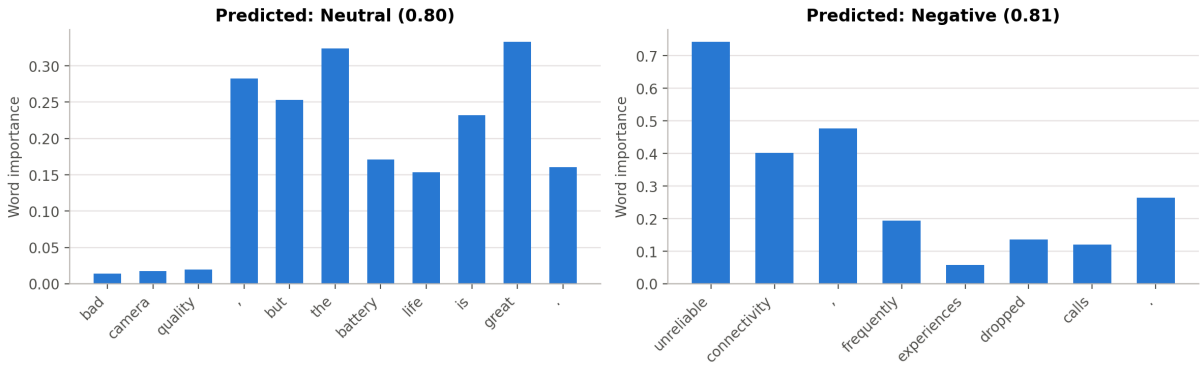


Figure 4.6: Integrated Gradients word importance for two sentences. A taller bar means that the word pushed the model more strongly toward its prediction.

Inference. In the dropped-calls review, the word "unreliable" receives the highest score (0.74), so the negative label rests on the same word a human reader would point to. In the mixed sentence, the decision is carried by the clause after "but", led by "great" (0.33), while "bad camera quality" scores close to zero. In other words, the model relied on the positive half of the sentence, which explains why it did not classify the review as negative.

Table 4.10 lists the 14 hard sentences together with the expected label and the predictions of both models.

Table 4.10: Hard sentences with the expected label and the predictions of the best base paper model and the proposed model.

Review	**Expected**	**Base paper (Logistic Regression)**	**Proposed**
Unreliable connectivity, frequently experiences dropped calls.	Negative	Negative	Negative
The battery is not good at all.	Negative	Neutral	Negative
Not bad for the price.	Positive	Neutral	Neutral
I didn't like the camera.	Negative	Neutral	Negative
I expected it to be terrible, but it works perfectly.	Positive	Positive	Positive
It is not worth the money.	Negative	Negative	Negative
Never had a single problem with this charger.	Positive	Neutral	Positive
The case is fine, nothing special.	Neutral	Neutral	Neutral
Would not recommend this to anyone.	Negative	Negative	Negative
Works as described, no complaints.	Positive	Positive	Positive
It stopped charging after two days.	Negative	Negative	Negative
Not the best case, but it does the job.	Neutral	Positive	Neutral
Great, another cable that broke in a week.	Negative	Negative	Positive
Hardly any battery drain overnight.	Positive	Negative	Positive
Correct		**8 of 14**	**12 of 14**

Inference. The proposed model corrects five sentences that the baseline misclassified, four of which involve negation, such as "not good at all" and "never had a single problem". Neither model handles "Not bad for the price" correctly. The proposed model also reads the sarcastic "Great, another cable that broke in a week" as positive, a case the baseline gets right, so sarcasm remains a clear weakness.

In bulk mode, the model added sentiment and emotion labels to all 11,357 test reviews in 22.6 seconds (502 reviews per second) and extracted aspects in a further 55.1 seconds. The overall throughput of 146 reviews per second on the T4 GPU means that a file of 100,000 reviews can be processed in about 11 minutes. Figure 4.7 shows the dashboard generated for this file.

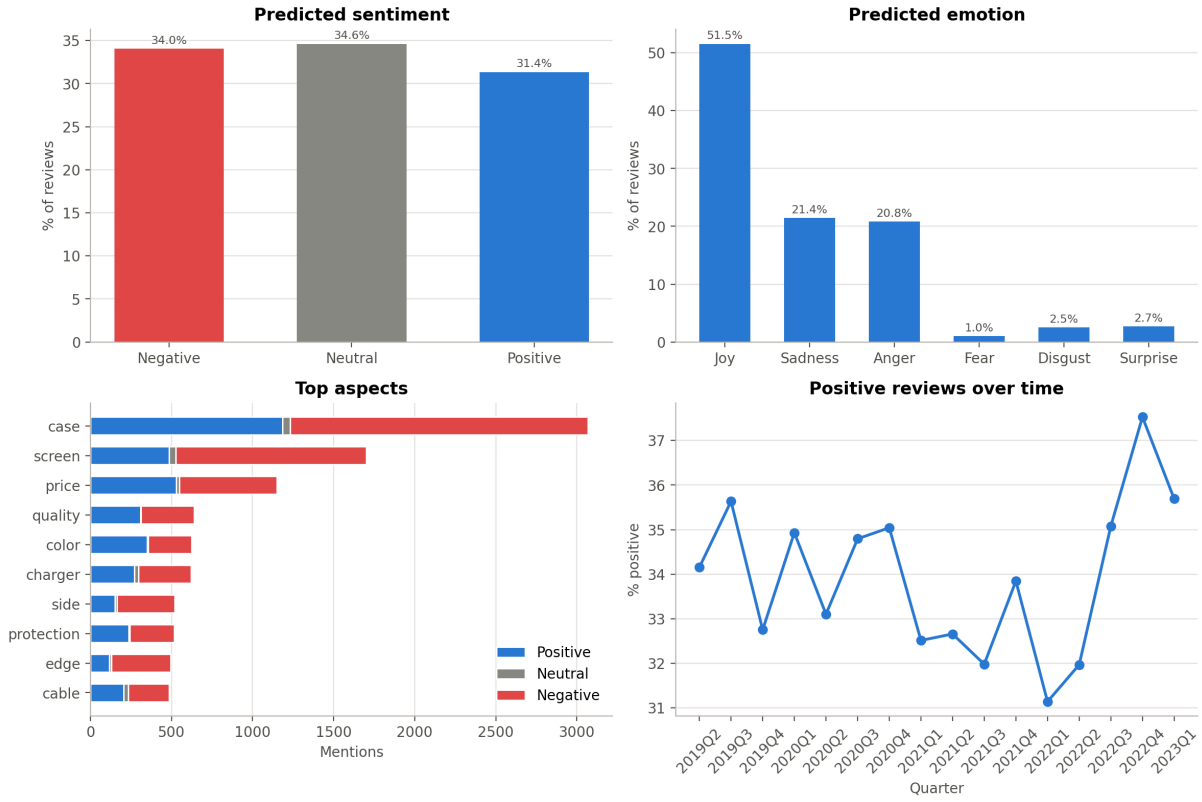


Figure 4.7: Dashboard produced for the labelled test file.

Inference. The predicted split (34.0% negative, 34.6% neutral and 31.4% positive) is close to the true split of the file (34.1%, 33.8% and 32.0%), so the bulk labels give a faithful picture of the file as a whole. Joy, sadness and anger together account for 94% of reviews, while fear, disgust and surprise are rare in product reviews. The share of positive reviews varied between 31% and 38% per quarter from 2019 to early 2023. Aspect extraction with spaCy accounts for 71% of the bulk processing time and is therefore the first step to optimise.

4.6 Demo Application

To make the system easy to demonstrate, a web interface was built with Gradio. It runs the trained model from the same Colab notebook and provides a public link that opens in any browser. The interface has two tabs, one for analysing a single review and one for analysing a complete CSV file, which are shown in Figures 4.8 and 4.9.

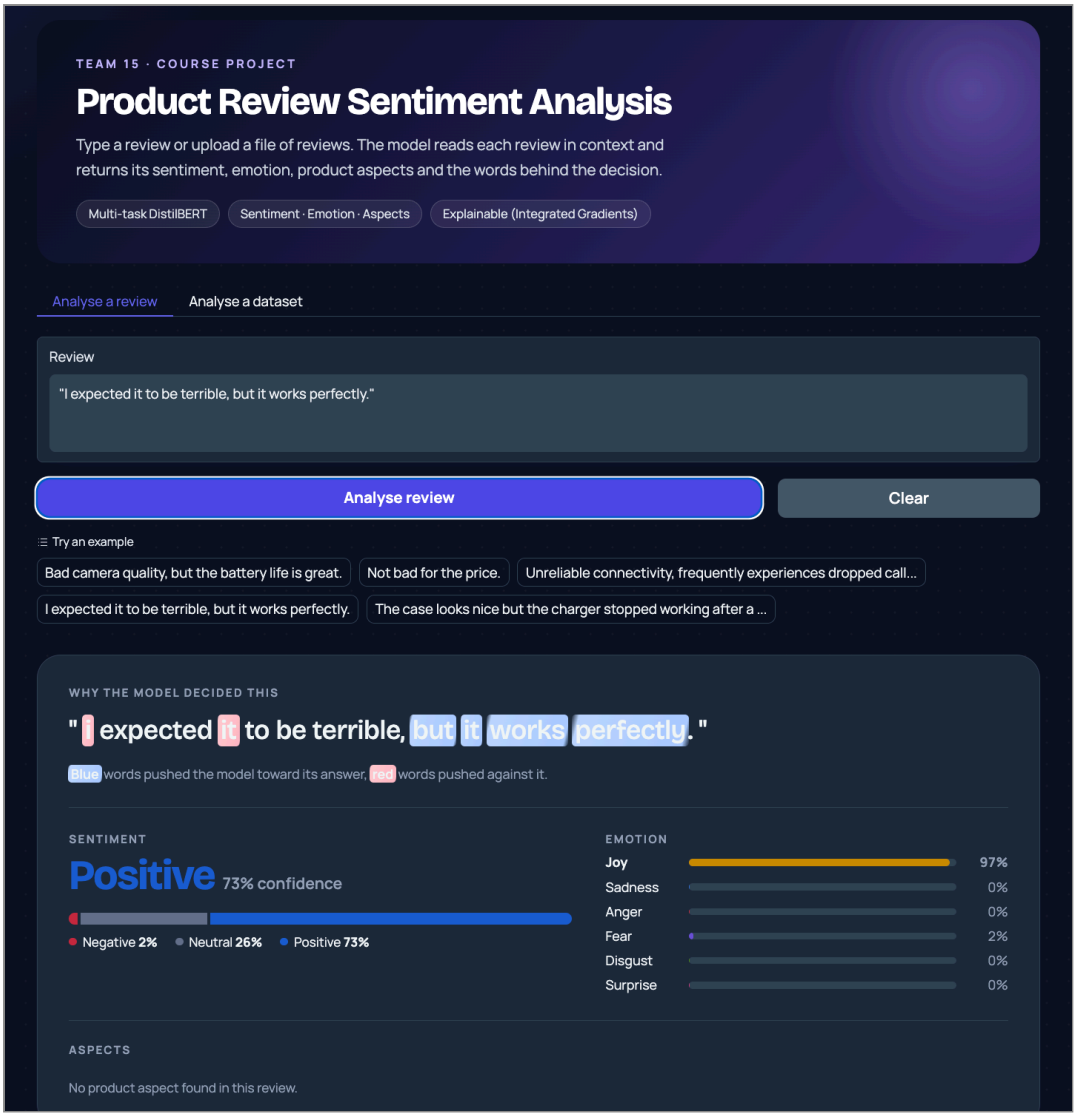


Figure 4.8: Web application, single-review tab. "I expected it to be terrible, but it works perfectly." is classified Positive (73%) with joy as the emotion (97%). Blue highlights mark the words that pushed the model toward its answer.

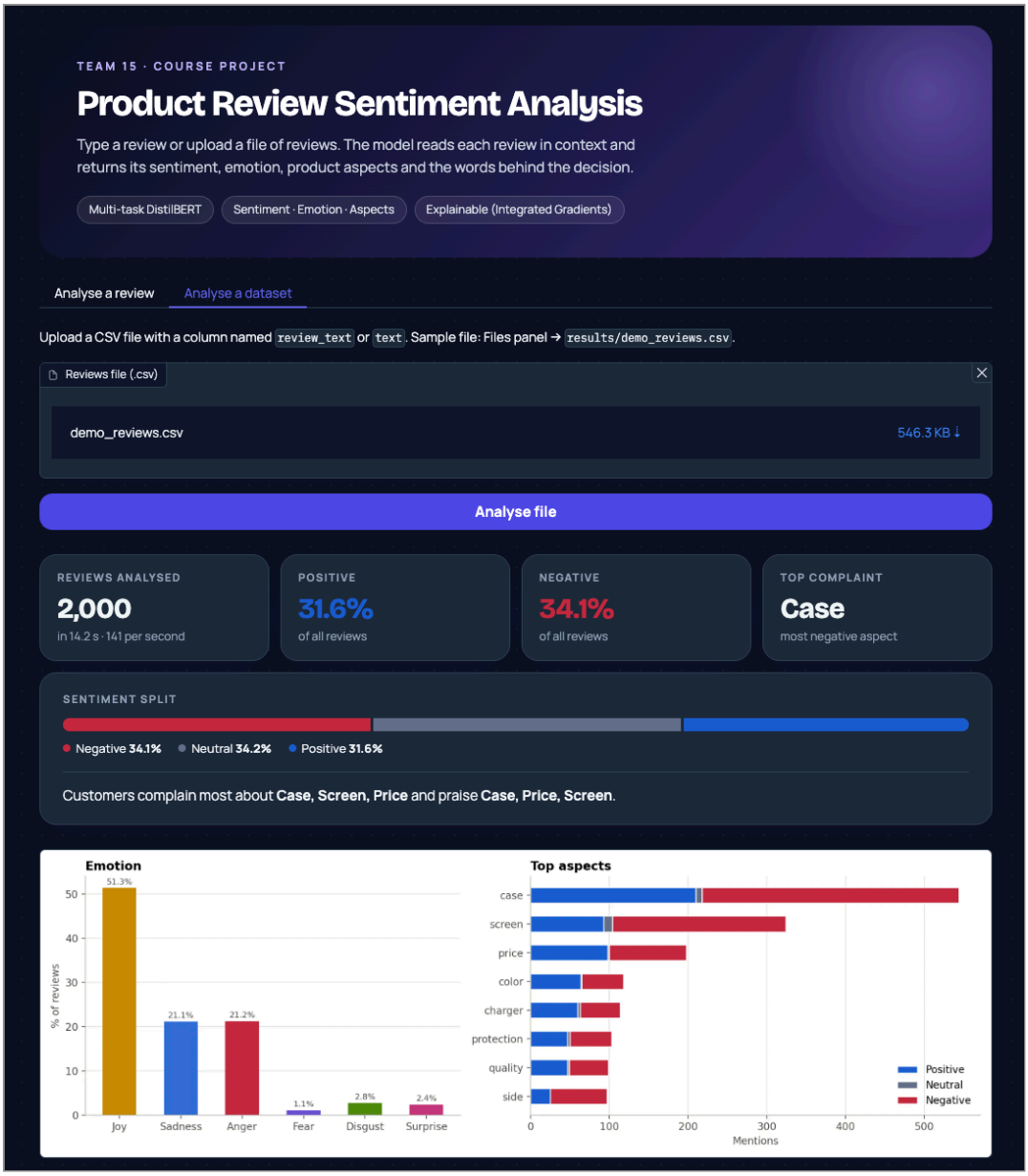


Figure 4.9: Web application, dataset tab after uploading 2,000 test reviews: summary boxes, sentiment split, emotion chart and top aspects.

Inference. The interface handles the contrast sentence correctly: the word "terrible" in the first half does not mislead the model, and the highlights fall on "but it works perfectly". In the dataset tab, 2,000 reviews were labelled in 14.2 seconds (141 per second), close to the 146 reviews per second measured in Section 4.5. The resulting split (34.1% negative, 34.2% neutral and 31.6% positive), and the identification of the case as the top complaint, are consistent with the results for the full test file.

5. PERFORMANCE AND METRICS

On the same 11,357 test reviews, the proposed model reaches an accuracy of 76.1% and a macro-F1 score of 0.764. The strongest base paper model, Logistic Regression, reaches 68.3% and 0.684, so the proposed model is 7.9 points higher in accuracy and 8.0 points higher in macro-F1. Because the base paper's models were re-run with their original preprocessing on this data, all of the figures in this section come from the same test set.

5.1 Evaluation Metrics

For each class, TP, FP and FN denote the true positives, false positives and false negatives on the test set. The metrics are defined as follows:

- **Accuracy** = correct predictions ÷ total predictions
- **Precision** = TP ÷ (TP + FP)
- **Recall** = TP ÷ (TP + FN)
- **F1** = 2 × Precision × Recall ÷ (Precision + Recall)
- **Macro-F1** = (F1 of Negative + F1 of Neutral + F1 of Positive) ÷ 3

Macro-F1 is used as the main metric because it gives equal weight to the Negative, Neutral and Positive classes, so a model cannot score well simply by favouring the largest class. Training time and bulk-mode throughput (reviews per second) are reported as well.

5.2 Models Compared

Table 5.1 lists the seven models compared in this study.

Table 5.1: Models compared in this study.

Type	Model	Features
Existing (base paper [1])	Multinomial Naive Bayes	TF-IDF (unigrams and bigrams)
Existing (base paper [1])	Linear SVC	TF-IDF
Existing (base paper [1])	Logistic Regression	TF-IDF
Existing (base paper [1])	MLP	TF-IDF
Existing (base paper [1])	Random Forest	TF-IDF
Ablation	DistilBERT, sentiment only	Contextual embeddings

Type	Model	Features
Proposed	Multi-task DistilBERT (sentiment + emotion + aspect)	Contextual embeddings

The base paper's models use its own preprocessing, namely lowercasing, stop-word removal and lemmatisation. Naive Bayes and Linear SVC are tuned with three-fold grid search, while MLP and Random Forest are trained on a 30,000-review subset to keep the run time practical.

5.3 Comparison with the Base Paper

Table 5.2 reports the test-set results of all seven models, and Figure 5.1 shows the same accuracy and macro-F1 values side by side.

Table 5.2: Test-set results of all models on the same 11,357 reviews.

Model	Accuracy	Precision	Recall	Macro-F1	Training time (s)	Trained on (reviews)
Naive Bayes (base paper)	0.667	0.669	0.669	0.669	0.3	90,855
Linear SVC (base paper)	0.678	0.676	0.680	0.677	25.2	90,855
Logistic Regression (base paper)	0.683	0.684	0.684	0.684	13.6	90,855
MLP (base paper)	0.665	0.663	0.667	0.664	114.1	30,000
Random Forest (base paper)	0.629	0.627	0.630	0.628	86.0	30,000
DistilBERT, sentiment only (ablation)	0.766	0.768	0.767	0.768	593.5	90,855
Multi-task DistilBERT (proposed)	0.761	0.765	0.763	0.764	1,053.6	90,855

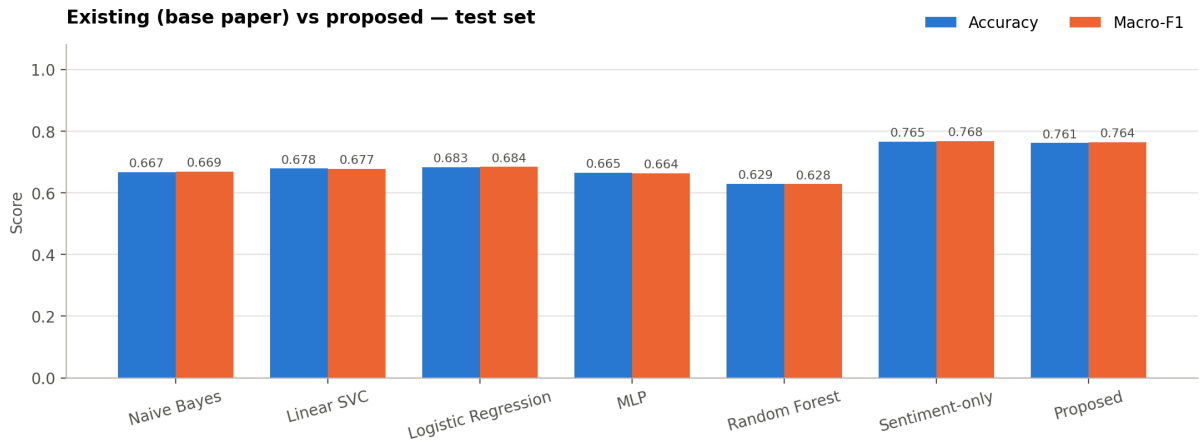


Figure 5.1: Accuracy and macro-F1 of the base paper's five models and the two DistilBERT models on the same test set.

Inference. Both DistilBERT models outperform every TF-IDF model by at least 7.8 points of accuracy. Among the base paper's models, Logistic Regression performs best and Random Forest worst (62.9%). The improvement does come at a cost in training time: about 18 minutes on a GPU, compared with under two minutes for any of the TF-IDF models.

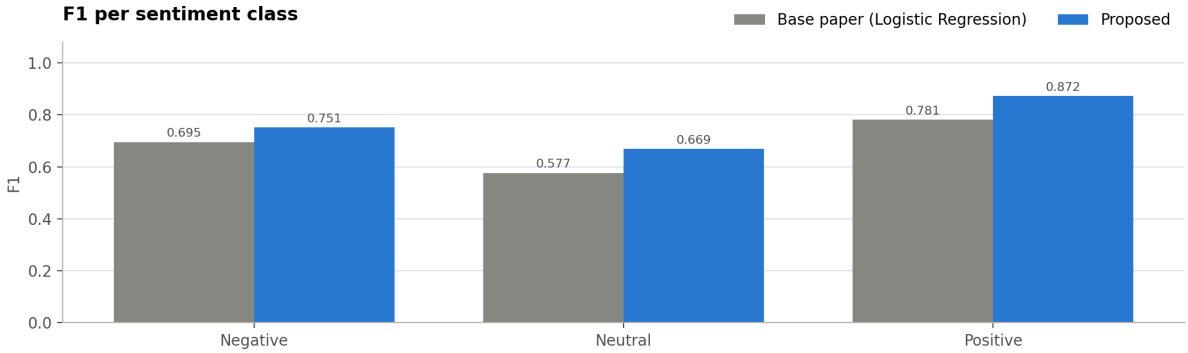


Figure 5.2: F1 for each sentiment class, best base paper model against the proposed model.

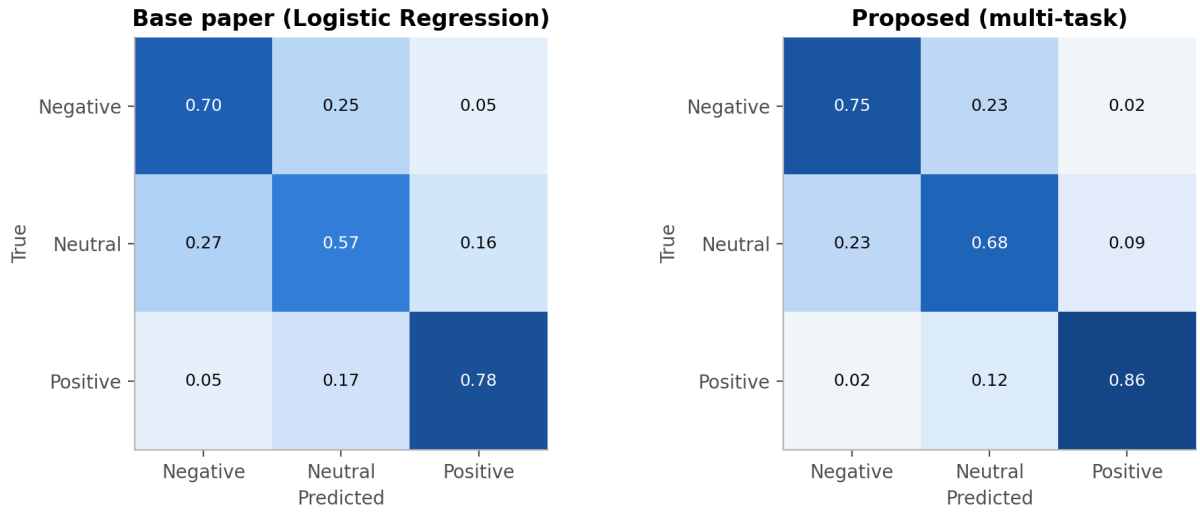


Figure 5.3: Confusion matrices of the best base paper model and the proposed model on the same test reviews.

Inference. The proposed model performs better on every class. The largest F1 gains are on neutral reviews (0.577 to 0.669) and positive reviews (0.781 to 0.872), with a smaller gain on negative reviews (0.695 to 0.751). The share of correct predictions rises from 70% to 75% for negative reviews, from 57% to 68% for neutral reviews and from 78% to 86% for positive reviews. Serious errors, in which positive and negative are confused, fall from 5% to 2%.

5.4 Overall Comparison

Table 5.3 brings together the main differences between the base paper's method and the proposed system.

Table 5.3: Overall comparison of the base paper method and the proposed system.

Measure	Base paper method	Proposed system
Sentiment accuracy (11,357 test reviews)	68.3% (Logistic Regression)	76.1%
Sentiment macro-F1	0.684	0.764
Hard sentences correct (negation, contrast, sarcasm)	8 of 14	12 of 14
Emotion	Separate model	Same model, macro-F1 0.700 on GoEmotions
Aspect-level sentiment	Not available	Macro-F1 0.712 on SemEval-2014
Explanation of each prediction	Not available	Top words by Integrated Gradients

Overall, the proposed system is more accurate on the base paper's own task and adds three capabilities that the base paper does not offer. Its main limitations lie in neutral reviews, sarcasm, and aspect sentiment in sentences that contrast two parts.

6. CONCLUSION

This project set out to extend the base paper on Amazon product review sentiment analysis, and it does so in five ways. A fine-tuned DistilBERT encoder replaces the TF-IDF features, allowing words such as "not" and "but" to be interpreted in context. A single multi-task model predicts sentiment and emotion together instead of relying on two separate models. An aspect module gives a separate opinion for each product part, such as the camera, battery or price, and Integrated Gradients highlights the words behind every prediction. Finally, the system accepts both a single sentence and a large review file.

The complete pipeline, including the base paper's models and the evaluation, runs in a single Colab notebook in about 40 minutes on a free T4 GPU. On 11,357 test reviews, the proposed model reached an accuracy of 76.1% and a macro-F1 score of 0.764, which is 7.9 and 8.0 points above the best base paper model, and it correctly classified 12 of 14 hard sentences, compared with 8 for the baseline. The same encoder also delivers emotion detection (macro-F1 0.700) and aspect-level sentiment (macro-F1 0.712), while staying within 0.4 points of a dedicated sentiment-only model. The main weaknesses that remain are neutral reviews, sarcasm, and aspect sentiment in sentences that contrast two parts.

Future work. Several directions follow naturally from these results:

- Training the aspect head on phone and accessory reviews, since the 1,769 laptop pairs used here are few in number and come from a different domain.
- Improving the handling of sarcasm with dedicated sarcasm data.
- Evaluating larger encoders such as RoBERTa [17] to improve accuracy on neutral reviews.
- Learning aspects automatically rather than from a frequency list.
- Supporting Indian languages and mixed-language reviews such as Hinglish and Tanglish.
- Deploying the system as a web service for continuous review monitoring.

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